Kaya

A multi-organ digital twin for Type 2 Diabetes

It shows what glucose does to the liver, eyes and kidneys, and lets a doctor test a change first.

Team Kaya · IIT Hyderabad · Dulari Bhushan, Prateek Giddaplavar

Happiest Health Reimagining and Reforming Healthcare in India Summit 2026 · Digital Twin Challenge, Phase 1

The damage from diabetes is silent

Glucose is tracked. The organs it harms are checked once a year, if at all.

100M+

people with diabetes in India (ICMR-INDIAB)

21.7%

of Indians with diabetes had retinopathy (AIOS screening)

22%

of US adults with diabetes have advanced liver stiffness (NHANES)

- Fatty liver, retinopathy and kidney disease build for years before symptoms.
- Existing twins and apps forecast sugar and coach diet. None we reviewed links it to organ damage.

What exists, and the gap

Reviewed from public sources

Who	What it does	Gap
Twin Health	Remission program: CGM, wearables, care team	Reports liver and weight benefit; closed program
Fitterfly, sugar.fit, Ultrahuman	CGM plus coaching and scores for Indian users	Glucose and lifestyle; no organ outcomes
Dexcom, Roche SmartGuide	Predictive low-glucose alerts 20-30 min ahead	Glucose only
Remidio + RetinaRisk	Offline smartphone DR screening, risk-based interval	Eye only; no wearable link
ARTEMIs, HEPATWIN (EU)	Virtual organ twins for liver disease	Research stage; liver only
Kaya	One twin: glucose to liver and eye, plus what-if	**Open; validated per module**

What Kaya does

Two data streams fused into one twin, five questions answered, one place to test a change

Static: labs, genetics, history + Dynamic: CGM, heart rate, sleep, steps, meals

Glucose

What will glucose do in the next 2 hours?

Liver

Is fat building up? Is fibrosis developing?

Eye

What is the retinopathy grade and where is it heading?

Kidney

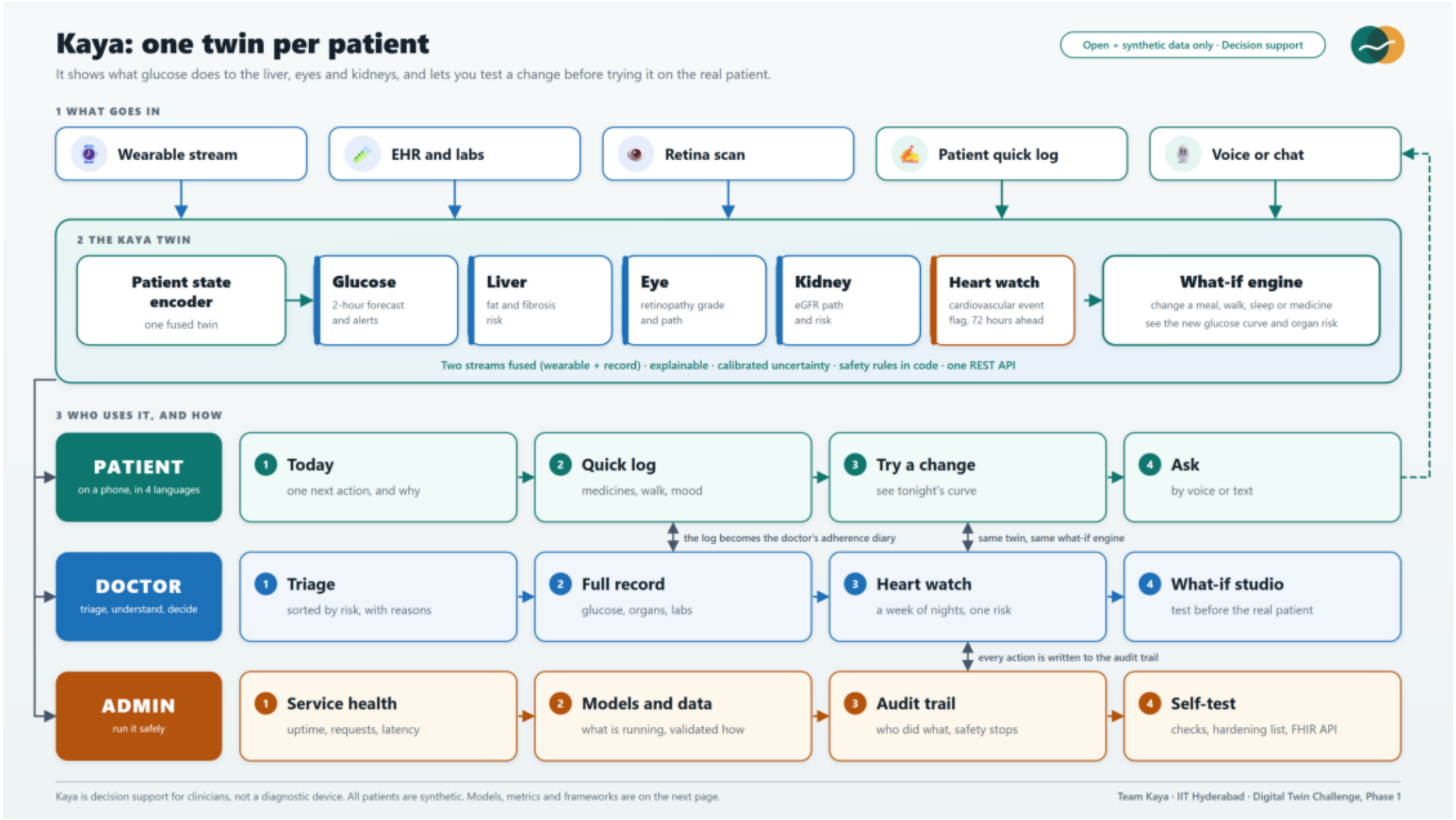
Is kidney function declining, and how fast?

Heart watch

Is a cardiovascular event coming in the next 72 hours?

What-if engine

Change carbohydrates, walking, medication or steps, and see the glucose curve and one-year organ risk move


Kaya: one twin per patient
It shows what glucose does to the liver, eyes and kidneys, and lets you test a change before trying it on the real patient.
Open + synthetic data only · Decision support
1 WHAT GOES IN
Wearable stream
EHR and labs
Retina scan
Patient quick log
Voice or chat
2 THE KAYA TWIN
Patient state encoder
one fused twin
Glucose
2-hour forecast and alerts
Liver
fat and fibrosis risk
Eye
retinopathy grade and path
Kidney
eGFR path and risk
Heart watch
cardiovascular event flag, 72 hours ahead
What-if engine
change a meal, walk, sleep or medicine
see the new glucose curve and organ risk
Two streams fused (wearable + record) · explainable · calibrated uncertainty · safety rules in code · one REST API
3 WHO USES IT, AND HOW
PATIENT
on a phone, in 4 languages
1 Today
one next action, and why
2 Quick log
medicines, walk, mood
3 Try a change
see tonight's curve
4 Ask
by voice or text
the log becomes the doctor's adherence diary
same twin, same what-if engine
DOCTOR
triage, understand, decide
1 Triage
sorted by risk, with reasons
2 Full record
glucose, organs, labs
3 Heart watch
a week of nights, one risk
4 What-if studio
test before the real patient
every action is written to the audit trail
ADMIN
run it safely
1 Service health
uptime, requests, latency
2 Models and data
what is running, validated how
3 Audit trail
who did what, safety stops
4 Self-test
checks, hardening list, FHIR API
Kaya is decision support for clinicians, not a diagnostic device. All patients are synthetic. Models, metrics and frameworks are on the next page.
Team Kaya · IIT Hyderabad · Digital Twin Challenge, Phase 1

Data: open or synthetic only

What is real, what is simulated, and how we validate

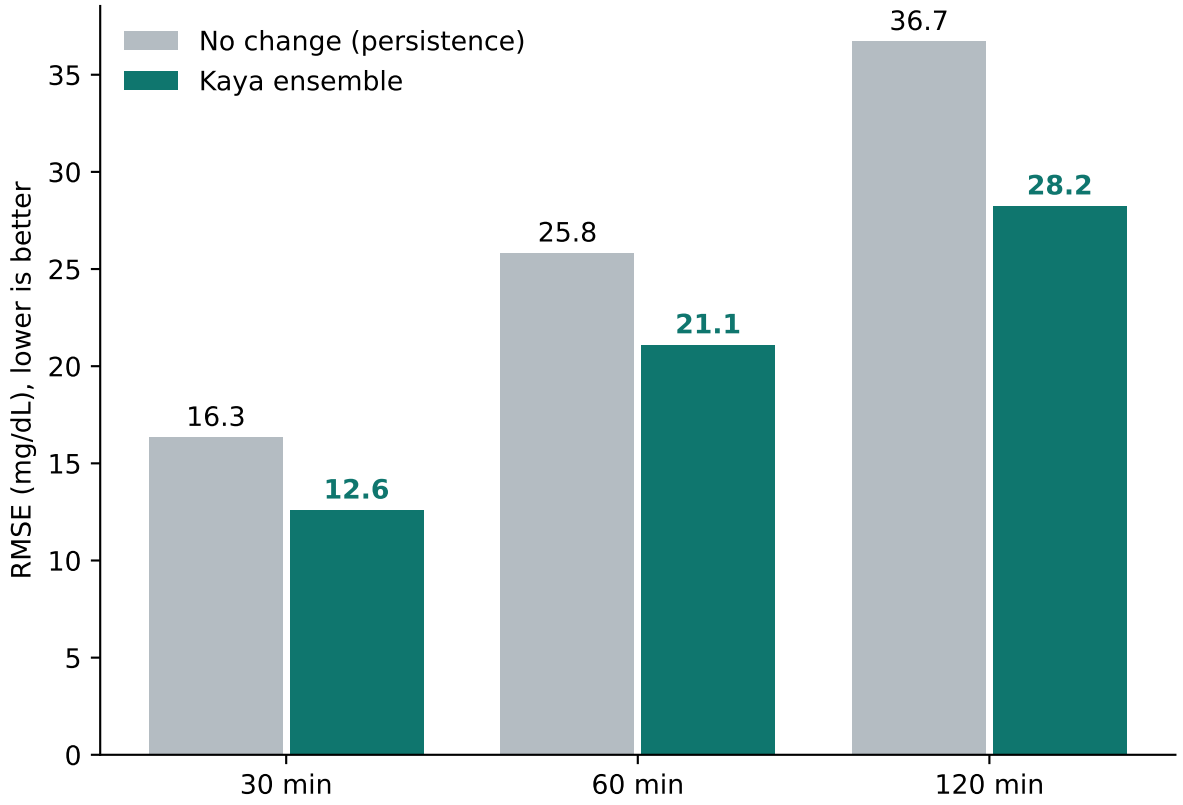
CGMacros (PhysioNet)	**Real**	Glucose forecaster: 34 subjects, subject-level cross-validation
NHANES 2017-2018	**Real**	Liver model: 4,745 adults with FibroScan labels
APTOS 2019 + IDRiD	**Real**	Retina grader: train on APTOS, test on Indian IDRiD
Synthetic cohort	**Simulated**	2,000 patients: links glucose patterns to organs for the twin and what-if
Coupled cohort (new)	**Simulated**	2,000 patients x 28 days: HRV, sleep and glucose coupled; cardiac events

Why this is honest

No public dataset links one person's glucose to liver, eye and kidney outcomes. Each module is checked on real data where it exists.
The glucose-to-organ links and what-if effect sizes are editable, literature-informed assumptions, labelled as simulation in the product.

Glucose forecasting on real, held-out subjects

CGMacros, 34 subjects, subject-level 5-fold cross-validation



72%

of low-glucose episodes caught

45 min

median warning before the low

80%

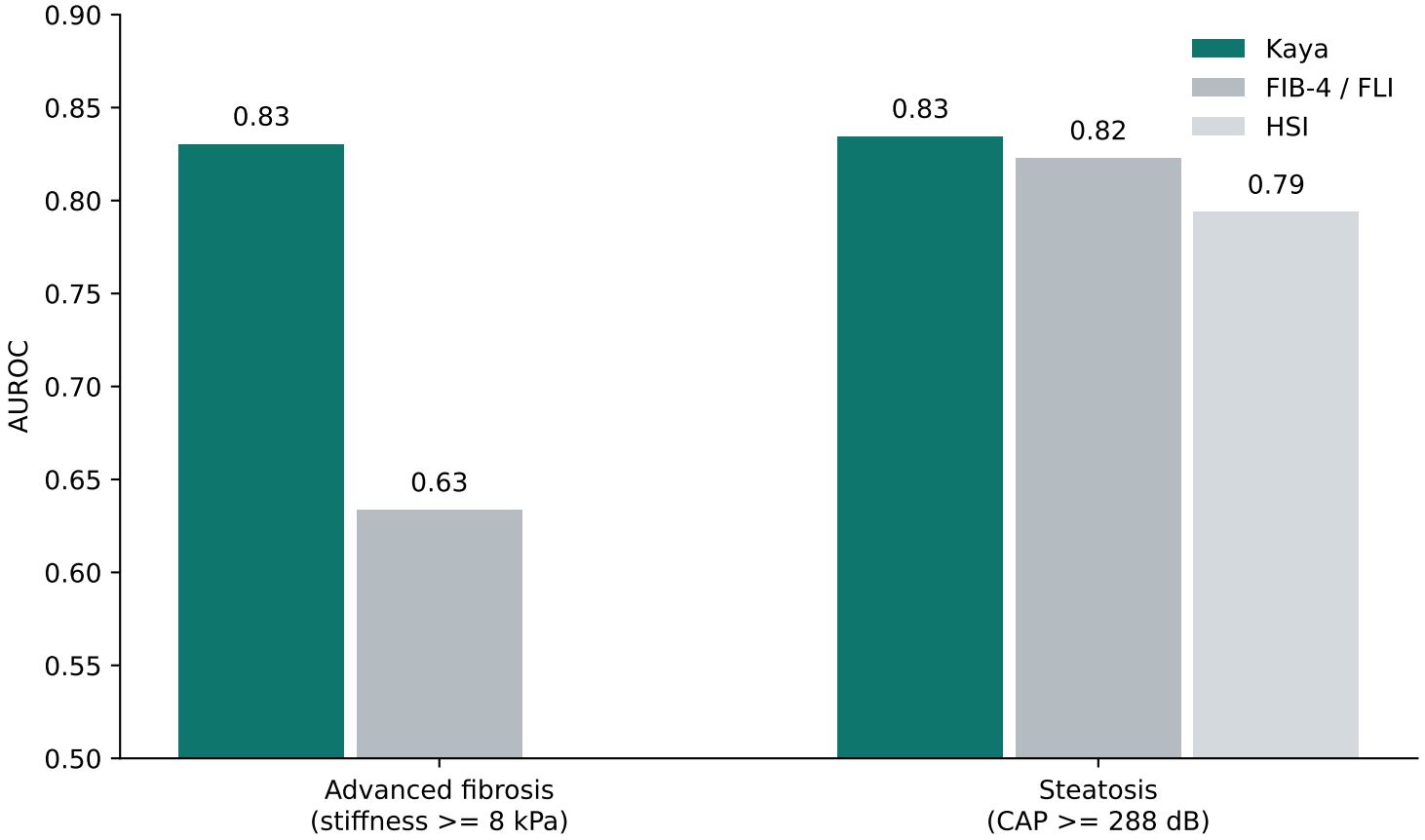
interval coverage (conformal, exact)

~1/day

false alerts per patient at that operating point

Liver: validated on real FibroScan data

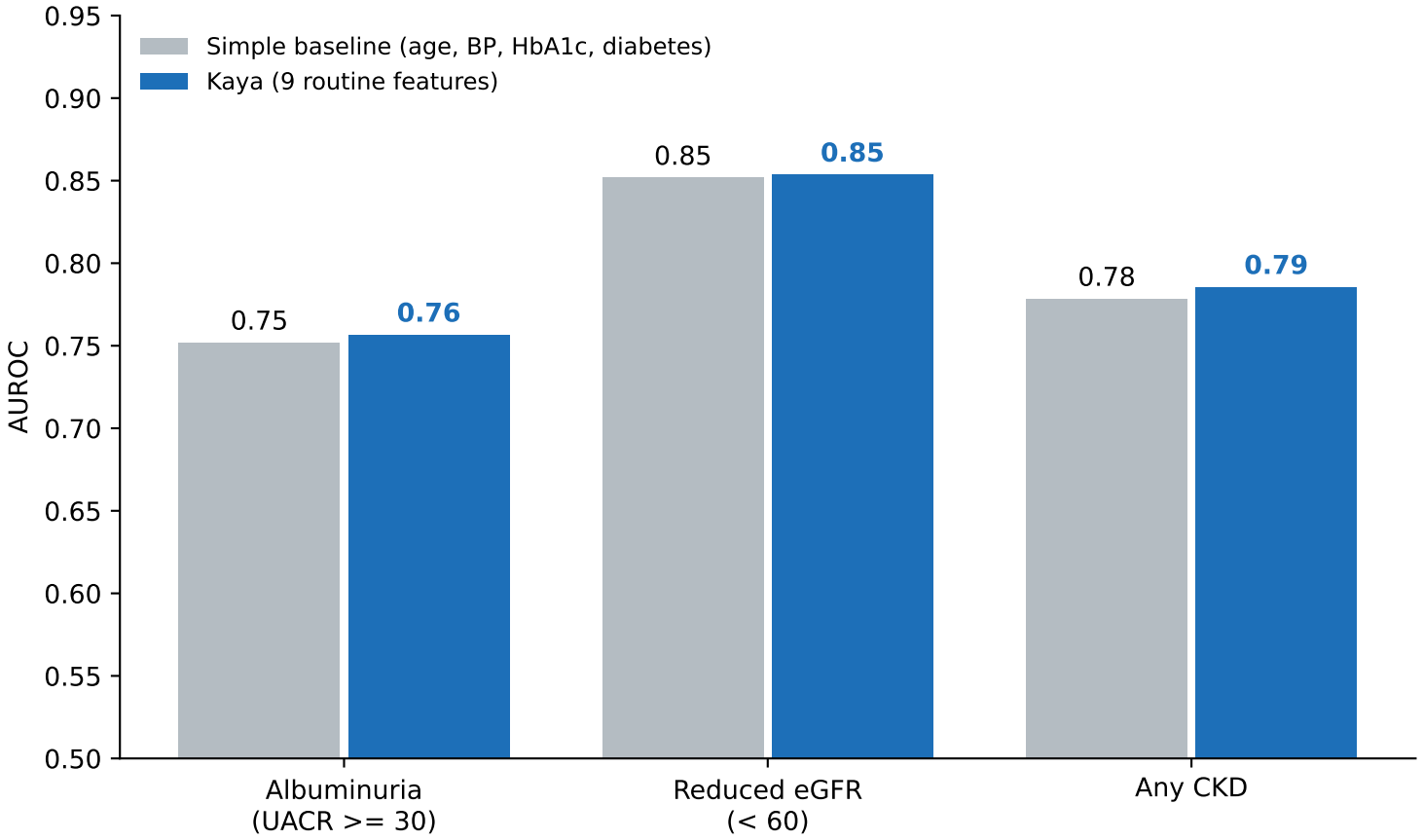
NHANES 2017-2018, 4,745 adults, 5-fold cross-validation, 95% CIs in the repository



- Fibrosis: 0.83 vs 0.63 for FIB-4
- Steatosis: 0.83 vs FLI 0.82, HSI 0.79
- Holds in adults with diabetes (0.76 / 0.80) and in A
- On top: a transparent glycemic-load state that mo

Kidney: real NHANES data, with honest limits

4,827 adults, 5-fold cross-validation; risk from routine measurements (no creatinine used)



- Reduced eGFR 0.85, albuminuria 0.76
- Adults with diabetes: 0.77 and 0.70
- Honest: only a small edge over a simple baseline, which is already strong
- On top: an eGFR path (about 1 to 4 mL/min per year) driven by HbA1c, BP, albuminuria and variability, as an editable simulation

Eye: retinopathy grading, tested on Indian images

EfficientNet-B2 trained on APTOS 2019, tested on IDRiD (Indian), ordinal grading

0.89

quadratic kappa
APTOS held-out

0.98

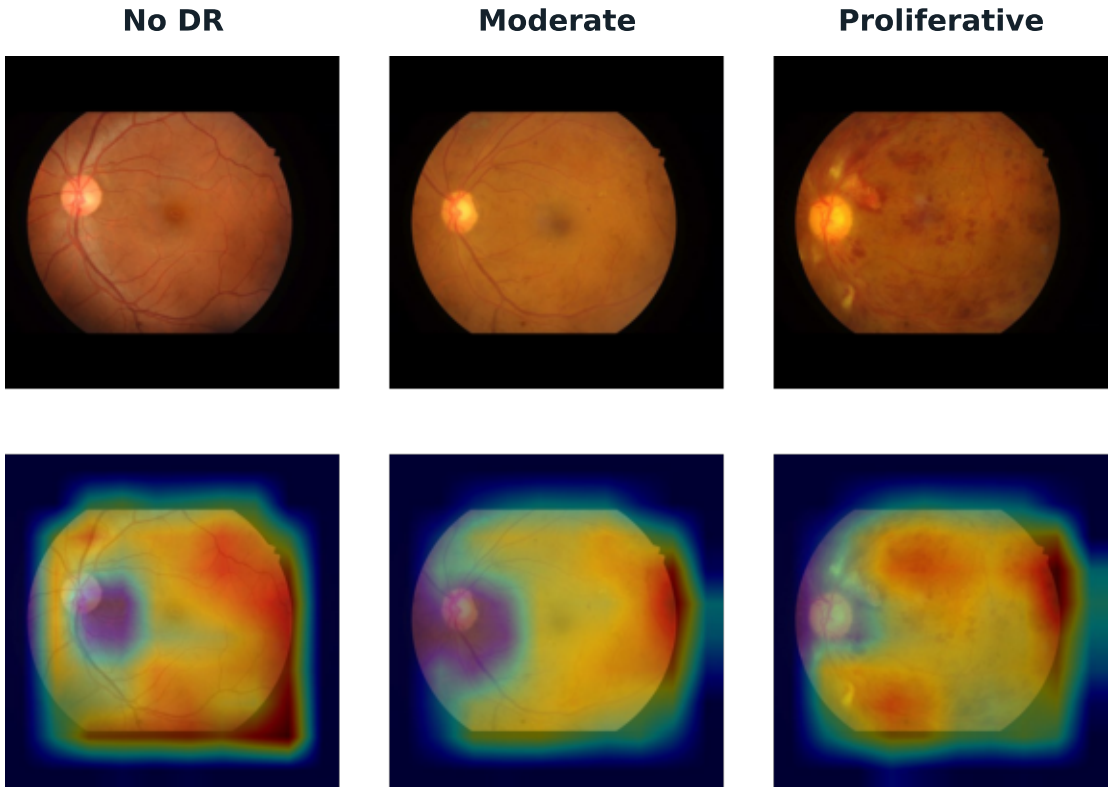
referable DR AUROC
APTOS held-out

0.71

quadratic kappa, IDRiD test
(Indian, fine-tuned locally)

0.92

referable DR AUROC
IDRiD test



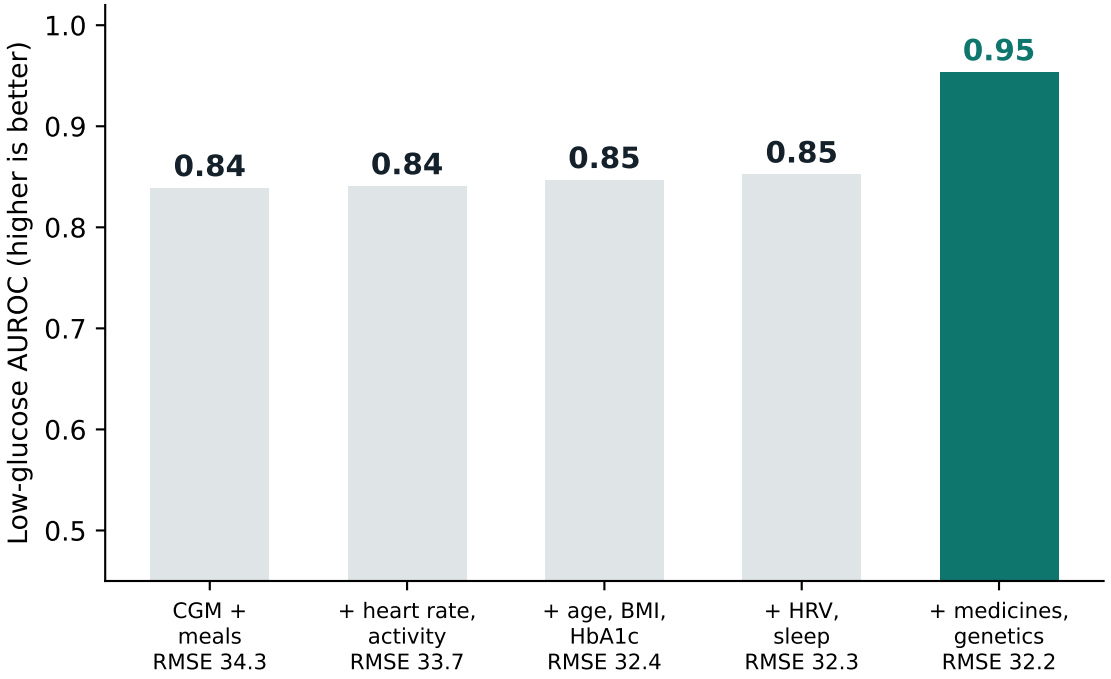
Top: IDRiD fundus. Bottom: Grad-CAM, where the model looked.

Without fine-tuning on IDRiD: kappa 0.57, AUROC 0.88. Camera and population shift is real, and local data closes part of the gap.

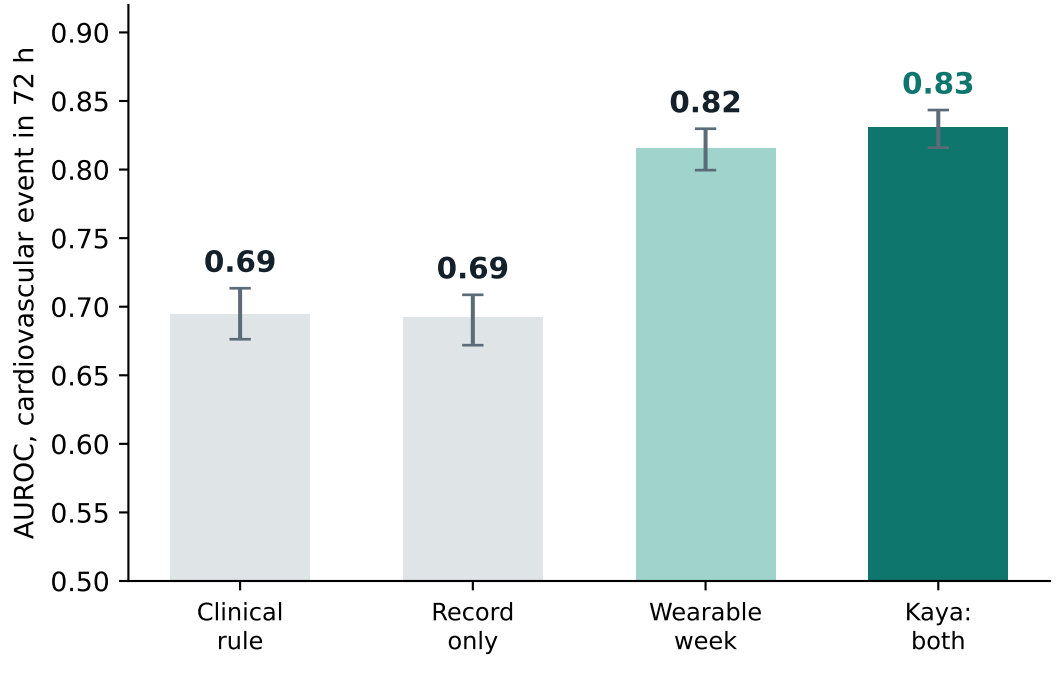
Two streams fused: what each one adds

Measured on held-out synthetic patients: input groups switched on one at a time (glucose) and a 5-fold study (heart)

Glucose: medicines lift low-glucose detection



Heart watch: the wearable week does most of the work



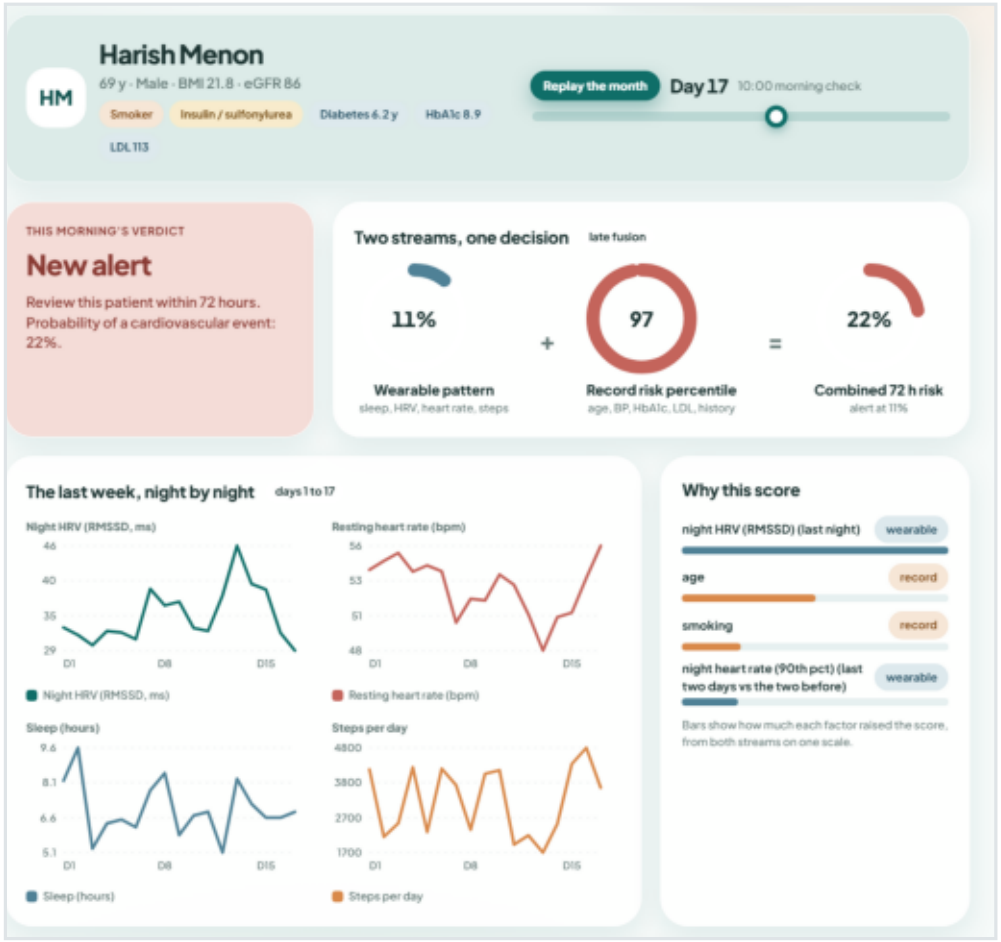
Heart watch at about 1 false alert per patient-month: 76% of events caught, median 23 h of warning (simple clinical rule: 44%).

For glucose, HRV and sleep add little (about 0.3 mg/dL at +2 h); the record, mainly medicines, adds the most. RMSE = forecast error at +2 h.

Synthetic cohort with assumed couplings and many more events than real life: it shows the fusion works as designed, not clinical accuracy.

Heart watch: a week of nights, read with the record

Each morning: wearable pattern + record risk = one 72-hour risk, with the reasons



Alert morning: HRV falling, resting heart rate rising

- The wearable model asks: is a prodrome under way?
- The record model asks: how dangerous is that for this person?
- A stacker combines the two; every alert lists its drivers
- Replay a month, then see what really happened, including a missed event and a false alarm
- 76% of events caught, median 23 h ahead, about 1 false alert per patient-month

Synthetic data. Decision support, not a diagnosis.

What-if: test the change on the virtual patient

Patient 1705, last 14 days replayed with three changes together

	Today	With the change
Time in range	42%	**71%**
Time above 180	58%	**29%**
Liver steatosis risk, 12 months	26%	**11%**
Advanced fibrosis, 12 months	35.5%	**24.4%**
Retinopathy progression, 2 years	6%	**2%**

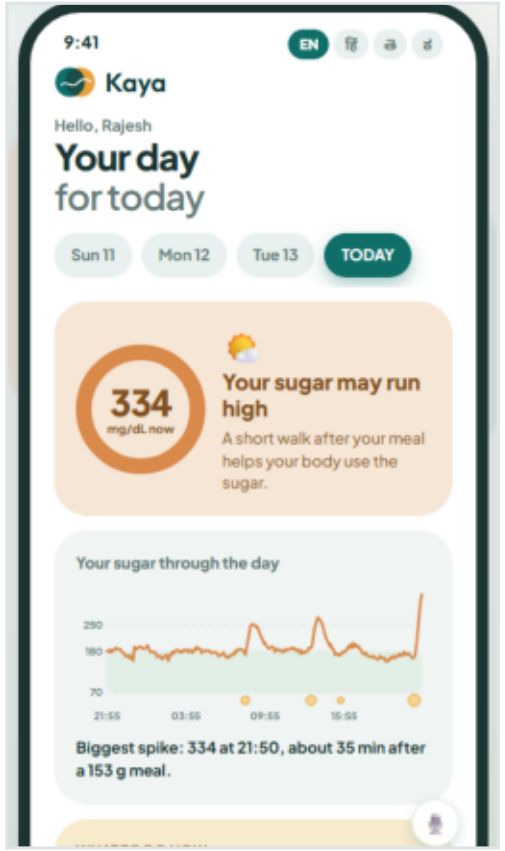
Walk 20 min after meals, 25% smaller carbohydrate portions, medication optimised. Effect sizes are editable, literature-informed assumptions and shown to the clinician as a simulation, not a prediction.

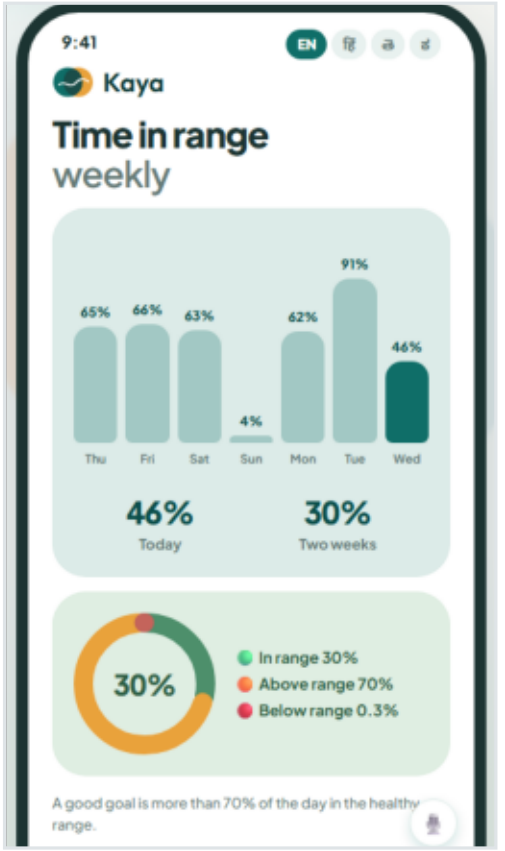
The product

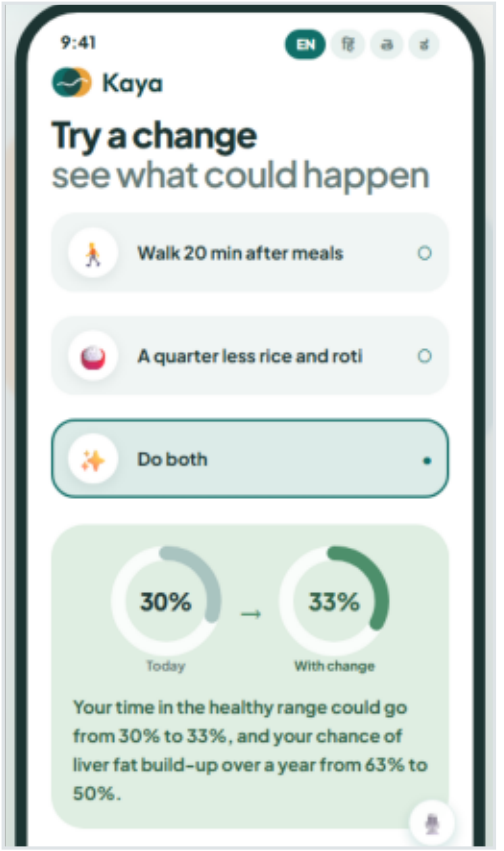
Doctor workspace and patient view. Replay shows the forecast against what actually happened.



Doctor workspace







Patient app: today, insights, try a change

Built like a product, not a notebook

Four features that make the twin usable in a clinic

Live mode

The glucose model runs on every incoming reading, about 1 ms each, and the dashboard updates in real time. Devices can POST readings to the API.

Explained

Why an alert fired, what drives each organ risk (SHAP), and where the retina model looked (Grad-CAM).

One-click report

A clinical summary PDF with the glucose chart, organ risks with their evidence level, a what-if table and recent alerts.

Voice and languages

Speak or type in English, Hindi, Telugu or Kannada; the app answers aloud. Safety rules run in code before the AI. Drafts pending native review.

Limitations, ethics and what is next

Limitations we state openly

- Glucose-to-organ links, what-if effect sizes and the Heart watch couplings are assumptions, not clinical findings
- Glucose model validated on 34 real subjects; patients in the demo are synthetic
- Decision support for clinicians, not a diagnostic device; not clinically validated

Roadmap

- An illustrative, clearly labelled retina progression preview and a blood-pressure what-if
- Consent records, audit log, role-based access, fuller FHIR and ABDM alignment
- Native-speaker review of the Hindi, Telugu and Kannada patient app; synthetic clinical notes with a text model
- Validate the glucose-to-organ links and the Heart watch on real longitudinal data when available

Existing twins manage sugar.

Kaya shows what sugar does to the organs, and lets a doctor test the fix first.

Open source, Apache-2.0 · open and synthetic data only · github: see README

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